

Phishing Email Detection Model

Project Overview

A machine learning project using Scikit-learn to detect whether an email is Phishing or Safe (Legitimate) based on email content and URL-related features.

This project builds an intelligent phishing email classifier using Natural Language Processing (NLP) and Machine Learning techniques.

The model:

- Reads email datasets
 - Extracts important textual and URL features
 - Trains a classifier using Scikit-learn
 - Predicts whether an email is phishing or legitimate
 - Displays evaluation metrics including:
 - Accuracy Score
 - Classification Report
 - Confusion Matrix
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Objectives

- Detect phishing emails automatically
 - Analyze suspicious URLs and keywords
 - Learn text preprocessing and ML classification
 - Improve cybersecurity awareness
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Technologies Used

- Python
 - Scikit-learn
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - NLTK
 - Regex
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Dataset

You can use datasets from:

- Kaggle phishing email datasets
- SpamAssassin dataset
- Enron email dataset

Dataset should contain:

- Email text/content
 - Label:
 - Phishing
 - Safe
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Features Extracted

The model analyzes:

- Suspicious URLs
 - Number of links
 - Presence of IP addresses
 - Keywords like:
 - urgent
 - verify
 - password
 - click here
 - login
 - Special characters
 - Email length
 - TF-IDF textual features
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Machine Learning Workflow

1. Data Collection

Load phishing and legitimate email datasets.

2. Data Preprocessing

- Convert text to lowercase
- Remove punctuation
- Remove stopwords
- Tokenization

3. Feature Extraction

- TF-IDF Vectorizer
- URL feature extraction

4. Model Training

Train models such as:

- Logistic Regression
- Naive Bayes
- Random Forest
- Support Vector Machine (SVM)

5. Evaluation

Measure:

- Accuracy
 - Precision
 - Recall
 - F1-score
 - Confusion Matrix
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Python Implementation

```
# Import libraries
import pandas as pd
import numpy as np
import re
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load dataset
data = pd.read_csv("emails.csv")

# Convert labels
data['label'] = data['label'].map({
    'Phishing': 1,
    'Safe': 0
})

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    data['text'],
    data['label'],
    test_size=0.2,
    random_state=42
)

# Create ML pipeline
model = Pipeline([
    ('tfidf', TfidfVectorizer(stop_words='english')),
    ('classifier', MultinomialNB())
])

# Train model
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy * 100:.2f}%")

# Classification report
print(classification_report(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

# Plot confusion matrix
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Safe', 'Phishing'],
            yticklabels=['Safe', 'Phishing'])
plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```

How to Run

Step 1: Install Required Libraries

```
pip install pandas numpy scikit-learn matplotlib seaborn nltk
```

Step 2: Prepare Dataset

Save dataset as:
emails.csv

Required columns:
- text
- label

Step 3: Run the Program

```
python phishing_detector.py
```

Expected Output

Accuracy: 96.45%

Classification Report:
precision recall f1-score support

Safe 0.95 0.97 0.96
Phishing 0.97 0.95 0.96

Confusion matrix graph will also be displayed.

Cybersecurity Concepts Learned

- Email phishing attacks
 - NLP for cybersecurity
 - Feature engineering
 - Text classification
 - Threat detection using ML
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Future Improvements

- Real-time email scanning
 - Browser extension integration
 - Deep learning models
 - URL reputation checking
 - Email header analysis
 - Deployment using Flask or Streamlit
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Suggested Project Structure

Phishing-Email-Detection/

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■■■ dataset/

■ ■■■ emails.csv

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■■■ phishing_detector.py

■■■ requirements.txt

■■■ README.md

■■■ screenshots/

Expected Outcome

The model successfully classifies emails as:

- Phishing
- Safe

with high accuracy using:

- textual analysis
- keyword detection
- URL-based feature extraction
- machine learning algorithms.